

VIT-AP
UNIVERSITY

CAPSTONE PROJECT REVIEW-3



TITLE : Political Sentiment Analysis

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ABSTRACT

- Due to the drastically expanding user base of social media platforms, massive volumes of unstructured text are being created in the form of messages, chats, posts, and blogs.
- Social media, in addition to being a means of information exchange, is a surprisingly simple platform for expressing ideas and opinions that gain popularity when appreciated by a large number of people.
- This degree of acceptance might be an indicator of how people feel about a certain thing, person, or place. Social media platforms like Twitter generate enormous amounts of text containing political insights that may be mined to analyse voter emotion and predict future election trends.
- In this work, an attempt is made to mine tweets, capture the political sentiments from it and model it as a supervised learning problem.



OBJECTIVE

- To understand public opinion on a particular political issue or candidate.
- To predict election outcomes or measure the effectiveness of campaign messaging.
- To identify and track trends in public sentiment over time.
- To identify and analyze the language and sentiment of political discourse, including propaganda and misinformation.
- To inform and influence public policy by providing data-driven insights on public sentiment.



MOTIVATION

- To understand public opinion on a particular political issue or candidate. By analyzing the sentiment of public discourse, we can get a better understanding of how people feel about a particular issue or candidate. This can be useful for campaign strategists, policy makers, and other decision makers.
- To predict election outcomes or measure the effectiveness of campaign messaging. By analyzing public sentiment, we can get a sense of how people are likely to vote in an election or how effective a particular campaign message is. This can be useful for political strategists and analysts.
- To identify and track trends in public sentiment over time. By analyzing sentiment data over time, we can identify trends in public opinion and see how sentiment changes in response to events or campaigns.



MOTIVATION

- To identify and analyze the language and sentiment of political discourse, including propaganda and misinformation. By analyzing the sentiment and language of political discourse, we can identify and understand propaganda and misinformation, and take steps to counter it.
- To inform and influence public policy by providing data-driven insights on public sentiment. By understanding public sentiment, policy makers can make more informed decisions that are more likely to be supported by the public.



INTRODUCTION

- Twitter has been a popular platform in recent years for people to express their political views and engage with everyday people. The educated class, politicians, actors, and other celebrities are among the Twitter users, despite the fact that this percentage of India's population is quite tiny.
- These "influencers" have the power to use their tweets to influence the feelings and beliefs of their followers. It would be beneficial to be able to evaluate sentiment in order to divide material into categories and make it simpler for users to locate and respond to opposing viewpoints, improving the ability to exchange and discuss information.



INTRODUCTION

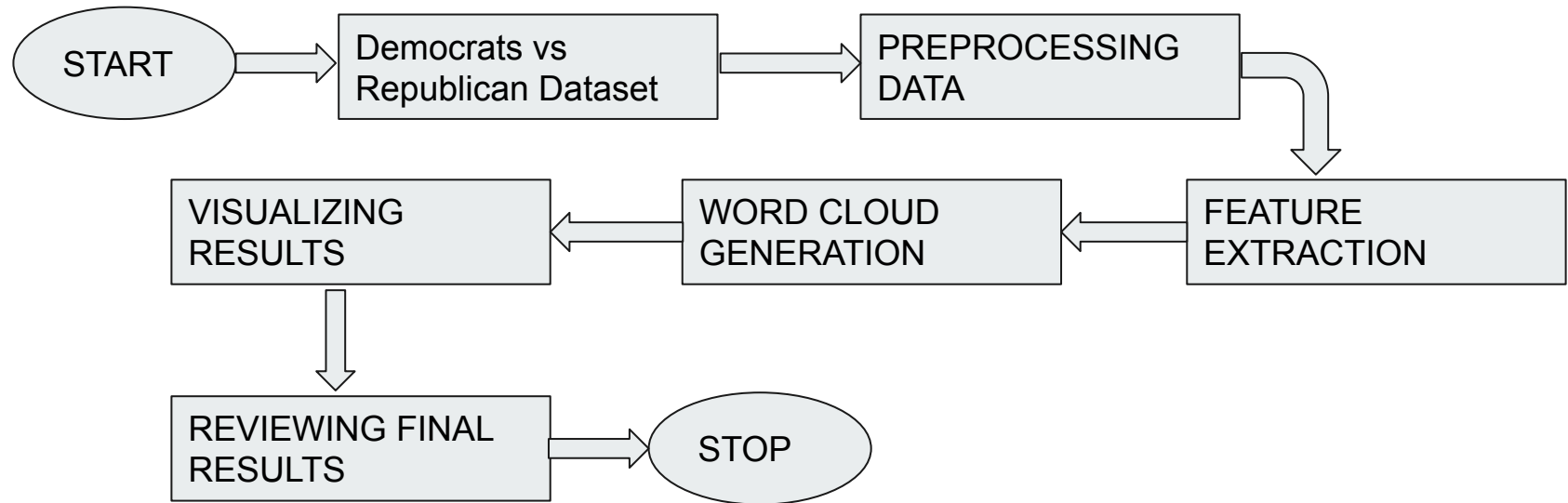
- By using the information available in the twitter regarding the political opinions we want to try to apply Sentiment Analysis. By using the NLP models and various other models we can get the results.
- This analysis would help the organizations who help the political parties with the idea whos going to support which party.

LITERATURE STUDY



S.no	Title of the Paper	Authors	Journal Details
1	Analysis of Political Sentiment Orientations on Twitter	Mohd Zeeshan Ansaria M.B.Aziz,M.O.Siddiqui ,H.Mehraa,K.P.Singha	International Conference on Computational Intelligence and Data Science (ICCIDS 2019)
2	POLITICAL SENTIMENT ANALYSIS ON DELHI USING MACHINE LEARNING	DHARMINDER YADAV , AVINASH SHARMA, SHARIK AHMAD, AND UMESH CHANDRA	JOURNALAdvances in Mathematics: Scientific Journal 9(2020),
3	Using sentiment analysis to define twitter political users' classes and their homophily during the 2016 American presidential election	Josemar A. Caetano, Hélder S. Lima, Mateus F. Santos & Humberto T. Marques-Neto	Journal of Internet Services and Applications volume 9, Article number: 18 (2018)
4	Political Sentiment Analysis through Social Media	Saurabh Dorle; Nitin Pise	2018 Second International Conference on Computing Methodologies and Communication (ICCMC)

PROPOSED MODEL





METHODOLOGY

1. Preprocessing the data: This includes cleaning and preparing the data for analysis. This may involve removing unnecessary characters, handling missing values, and standardizing the text.
2. Feature Extraction: This includes finding the subjectivity , polarity, and classifying them.This can be achieved through TextBlob .
3. Sentiment analysis: This is the main step of the process, where the sentiment of the text is analyzed and classified as positive, negative, or neutral. TextBlob offers a variety of pre-trained classifiers and sentiment analysis models that can be used for this step.
4. Visualizing and interpreting the results: Once the sentiment has been analyzed, it is often helpful to visualize the results in order to understand and interpret them. This may involve creating graphs, charts, or tables to display the results.

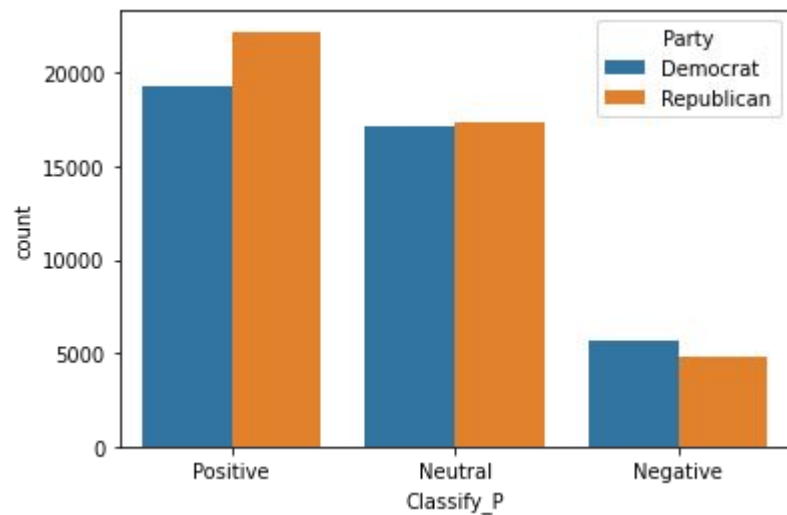


Implementation Details

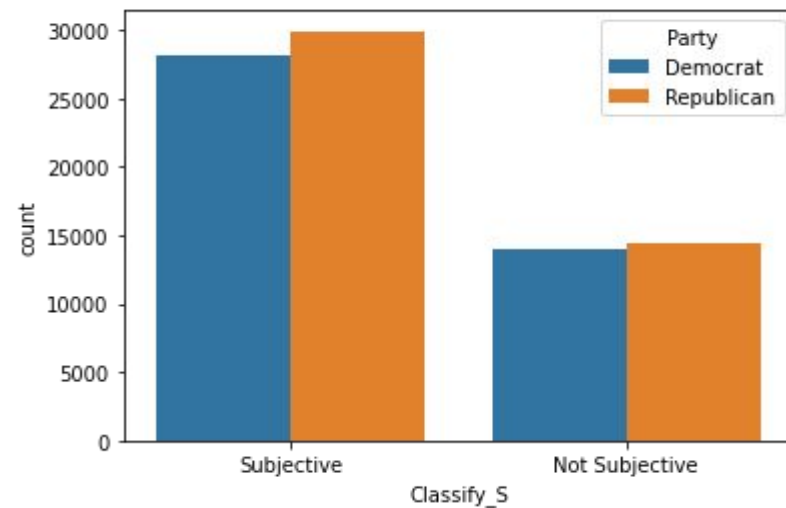
- TextBlob is a Python library for performing natural language processing (NLP) tasks. It is built on top of the popular NLP library NLTK (Natural Language Toolkit) and offers a simple and intuitive interface for a variety of NLP tasks, such as sentiment analysis, part-of-speech tagging, and language translation.
- With TextBlob, users can easily classify text as positive, negative, or neutral, as well as extract features such as the subjectivity and polarity of a text. TextBlob also offers pre-trained classifiers and sentiment analysis models, making it a user-friendly and efficient tool for conducting political sentiment analysis.
- By using TextBlob and other NLP tools, researchers and analysts can gain valuable insights into public opinion and better understand the sentiment of political discourse.



CLASSIFICATION OF POLARITY BETWEEN DEMOCRATS AND REPUBLICANS



CLASSIFICATION OF SUBJECTIVITY BETWEEN DEMOCRATS AND REPUBLICANS





AVERAGE POLARITY AND SUBJECTIVITY OF DEMOCRATS VS REPUBLICANS

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↳ Average Democrat Sentiment: 0.14, Democrat Subjective Tweet Percentage: 66.78%  
Average Republican Sentiment: 0.17, Republican Subjective Tweet Percentage: 67.39%
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FUTURE WORKS

- Improving the accuracy and reliability of sentiment analysis techniques. Currently, sentiment analysis techniques can be prone to errors and biases, and there is ongoing research into ways to improve the accuracy and reliability of these techniques.
- Expanding the scope of sentiment analysis to include more languages and regions. Currently, much of the research and development in sentiment analysis has focused on English-language data. There is a need to expand the scope of sentiment analysis to include more languages and regions in order to better understand global public opinion.
- Developing new approaches for handling propaganda and misinformation. With the proliferation of fake news and propaganda, there is a need for new approaches to identifying and addressing these issues in political sentiment analysis.



FUTURE WORKS

- Incorporating more sources of data. Currently, sentiment analysis often relies on social media data as a primary source of information. There is a need to incorporate more diverse sources of data, such as news articles, surveys, and other sources of public discourse, in order to get a more comprehensive understanding of public sentiment.
- Exploring the potential applications of sentiment analysis in policy making and decision making. There is a need to explore the potential applications of sentiment analysis in policy making and decision making, and to understand the ethical considerations and challenges involved in using sentiment analysis in these contexts.



Refereneces

- <https://www.sciencedirect.com/science/article/pii/S1877050920306669>
- <https://www.researchgate.net/>
- <https://ieeexplore.ieee.org/abstract/document/8487879>

